

PEIYU LIU

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SUMMARY

Ph.D. candidate in Biostatistics at the University of Florida with **pharmaceutical industry internship experience** and 5+ years of quantitative research experience in clinical trials and epidemiology. Expertise in **Bayesian inference and hierarchical modeling**, including **MCMC**, MAP estimation, mixed-effects and latent-variable models, and **missing-data methods**. Experienced in clinical trial design, exposure-response analysis, **dose optimization**, and **protocol/SAP development**. Familiar with **CDISC workflows**, including **SDTM**, **ADaM**, and **TLGs**. Proficient in **R**, **SAS**, **Python**, **Julia**, and **MATLAB** for statistical analysis, simulation, and reproducible research.

EXPERIENCES

Biostatistics Intern

Remote, U.S.

Hematology & Cell Therapy Biostatistic | Bristol Myers Squibb

May 2026 - Aug 2026

- Developed an **RShiny CUS** integrating efficacy, safety, and tolerability endpoints for hematology/oncology **benefit-risk evaluation**, enabling **reproducible OBD assessment** and decision-making among clinicians and statisticians.
- Extended the CUS to **binary, continuous, and time-to-event endpoints (RMST, Royston-Parmar, AFT)**, to provide a **consistent evaluation framework** for heterogeneous oncology outcomes.
- Prototyped an **AI-assisted CDISC traceability agent** linking **ADaM** → **SDTM** → **raw data**, enabling clinicians, statisticians, and programmers to rapidly verify derivations and streamline **cross-functional review**.
- Evaluated FDA Project **Optimus**—aligned **dose-optimization designs** (PK-BOIN12, STEIN12) through **Phase I/II trial simulations** and implemented RShiny workflows to support OBD selection.
- Automated raw-to-SDTM/ADaM variable mapping with **CDISC conformance checks**, reducing manual data preparation and **standardizing inputs** for **endpoint-specific sample-size calculations**.

Biostatistician

Gainesville, U.S.

Biostatistics Core | Emerging Pathogens Institute | University of Florida

May 2025 - Present

- Led biostatistical support across **randomized trials, observational EHR, vaccine, and infectious-disease studies**, translating scientific questions into reproducible methods, simulations, and analyses.
- Analyzed pediatric RSV severity in observational **EHR data** using **survival analysis (Cox PH)** and multivariable regression; partnered with **clinicians and virologists** to interpret findings, contributing to a peer-reviewed publication.
- Conducted BEAST-based **Bayesian phylodynamic analyses** of whole-genome RSV sequences, including prior specification and MCMC diagnostics, supporting robust inference of viral evolutionary dynamics.
- Developed **posterior-sampling and bootstrap frameworks** for a dengue cluster-randomized trial, quantifying endpoint-classification uncertainty and robustness of estimated intervention effects to support trial inference.
- Built **Bayesian ODE and PK-adherence models** of HIV viral dynamics, using **MCMC calibration** to assess how variability in drug exposure and adherence contributes to viral blips.
- Released the open-source **R package vaxineR**, standardizing outbreak-risk estimation, vaccination simulations, and visualization supporting reproducible analyses for a **UF faculty white paper**.

Research Assistant

Gainesville, U.S.

Biostatistics Department | University of Florida

Aug 2023 - Present

Project: Trial Endpoint Optimization for Arboviral Vector-Control Interventions — First author

- Developed a **simulation-validated** mechanistic framework comparing serological, clinical, and entomological endpoints for cluster-randomized trials, linking biological exposure models with **endpoint-specific sample-size calculations**.
- Derived **endpoint-specific sample-size formulas**, demonstrating up to **10⁶** smaller sample sizes for serology endpoints in low-incidence settings, results led to a first-author publication and **ASTMH oral presentation**.

Project: Practical identifiability and Bayesian inference of exposure from longitudinal serology — First author

- Developed a **Bayesian latent-process** framework to reconstruct seasonal mosquito-exposure dynamics from **longitudinal serology**, jointly modeling **latent exposure histories**, antibody kinetics, and inter-individual heterogeneity.
- Characterized **practical identifiability** under identity-masked repeated sampling and developed **robust estimating-equation** and uncertainty procedures, using simulations and three Benin/Thailand cohorts to assess sensitivity to dependence, heterogeneity, and kinetic assumptions.

Project: Bayesian Household Transmission Modeling with Incomplete Data — First Author

- Extended a mechanistic household final-size model to incomplete households using **Bayesian data augmentation**, deriving **missing-member likelihoods** and simulation-validated recovery of within-household transmission under **MAR missingness**.
- Implemented and optimized Bayesian MCMC in **Julia**, achieving approximately **10× faster** inference with verified convergence, and collaborated with **CDC epidemiologists** to interpret household versus community transmission patterns.

Selected Methodological Research Project — Molecular Epidemiology

Faculty Supervisor: Ira Longini | University of Florida

Gainesville, U.S.

Spring 2026

- Implemented **survival and transmission-tree likelihoods** integrating phylogenetic and epidemiologic time-to-event data; benchmarked epidemiologic-only versus phylogeny-informed estimators against known transmission across **1,000 stochastic SIR simulations**.

Research Fellow and Lecturer

Math Department | Hunan Institute of Science and Technology

Changsha, China

Jun 2021 - Aug 2023

- Developed a **mechanistic impulsive ODE model of tumor-immune dynamics** under radioimmunotherapy and used bifurcation and sensitivity analyses to identify treatment-intensity and immune-activation thresholds associated with tumor control.
- Directed a **~\$7,500 funded research project** applying **statistical-learning methods** (EM, Lasso, and GAMs) and spatial analyses to multinational COVID-19 data, characterizing weather-transmission associations across 10 countries.

EDUCATION

School of Public Health & Health Professions, University of Florida

Ph.D. in Biostatistics, GPA: 3.9/4

Gainesville, U.S.

Aug 2023 - Present

- **Coursework:** *Biostatistics Methods, Computing, Statistical Machine Learning, Survival Analysis, Causal Inference, Data visualization, Longitudinal Analysis, Analytical methods for Infectious Diseases*

School of Mathematics and Statistics, Shaanxi Normal University

M.S. in Computational Mathematics, top 5%

Xian, China

May 2021

- **Coursework:** *Advanced Biomathematics, Stochastic Process, Stability, Qualitative Theory of Differential Equations*

School of Mathematics and Statistics, Shaanxi Normal University

B.S. in Statistics, top 5%

Xian, China

May 2017

- **Coursework:** *Probability Theory, Statistical Analysis and R, Mathematical Modeling, Practical Regression Analysis, Multivariate Statistical Analysis, Statistical Computing, Time Series Analysis*

APPLICATIONS/PACKAGES

R Package

- Peiyu Liu, Matt Hitchings, Ira Longini. Package “vaxineR” Statistical toolkit for vaccine-coverage analysis and outbreak-risk modeling. <https://cran.r-project.org/web/packages/vaxineR/>

Clinical Data Tools

- Clinical Data Trace — CDISC lineage tracing from an analysis value back through ADaM, SDTM, and raw layers: <https://agent-cdisc-datasets-trace.streamlit.app/>
- Sample Size Suite — CDISC-aligned dataset checks and sample size calculation across time-to-event, continuous, and count endpoints: <https://peiyuliu.shinyapps.io/clinical-data-ssize-suite/>

RShiny Apps:

- Oncology Dose Optimization (Clinical Utility Score framework, exposure-response modeling, PK-BOIN dose finding, STEIN): <https://peiyuliu.shinyapps.io/oncology-pkboin-ERcus/>
- Dengue Clinical Trial (serological vs. clinical endpoints): <https://peiyuliu.shinyapps.io/anpp-trial-comparison/>
- US Weather Analyses (ggplot2 visualization, K-means clustering, Machine Learning forecasting): <https://peiyuliu.shinyapps.io/us-weather-analysis/>
- SIR Model Estimation (Least Squares, Bootstrap, MCMC): <https://peiyul.shinyapps.io/sir-estimation-comparison/>

PUBLICATIONS & MANUSCRIPTS

Google Scholar: scholar.google.com/citations?user=O6KFoOUAAAAJ

Published

1. **Liu, P.**, Rong, L., Cummings, D., & Hitchings, M. (2026). Mosquito salivary protein antibodies as efficient endpoints for vector control trials: A mechanistic comparison with clinical and entomological outcomes. *Infectious Disease Modelling*, 11(4), 1368–1386.
2. **Liu, P.**, He, S., Rong, L., & Tang, S. (2020). The effect of control measures on COVID-19 transmission in Italy: Comparison with Guangdong province in China. *Infectious Diseases of Poverty*, 9(130), 1–13.
3. Santos, L. A., Castillo, A., Gustani-Buss, E., Hitchings, M. D. T., **Liu, P.**, Iovine, N., Cash, M. N., Estrada, J. L., de Moraes, L., Lemey, P., Lednicky, J., Morris, J. G., Jr., Ryan, K., & Salemi, M. (2026). Genomic diversity of human respiratory syncytial virus (HRSV) type A in a pediatric cohort in the United States after the onset of the COVID-19 pandemic. (2026). *Microorganisms*, 14(8), 1656.
4. Yan, Q., & **Liu, P.** (2023). Effects of climatic factors and control measures on the spread of COVID-19. *Journal of Zhejiang University (Science Edition)*, 50(2).
5. Xia, F., Xiao, Y., **Liu, P.**, & Cheke, R. (2020). Differences in how interventions coupled with effective reproduction numbers account for marked variations in COVID-19 epidemic outcomes. *Mathematical Biosciences and Engineering*, 17, 85–98.
6. Wang, G., Ding, W., Zhang, Z., & **Liu, P.** (2022). Qualitative analysis of a tumor immune model with impulsive control strategies at fixed time. *Journal of Hunan Institute of Science and Technology*, 35(3), 1–6.

Under Review

7. **Liu, P.**, et al. (2026). Practical identifiability and Bayesian inference of seasonal exposure from serology data. *Manuscript under review*.

In Preparation

8. **Liu, P.**, et al. Bayesian household transmission modeling with incomplete data. *Manuscript in preparation*.
9. Robledo, J., Izquierdo, G., et al., & **Liu, P.**. A multi-site field study demonstrates that integrated biosecurity delays Fusarium wilt of banana spread in multi-host landscapes. *Manuscript in preparation*.
10. Robledo, J., **Liu, P.**, Vallad, G., & Garrett, K. Filling the gaps between host availability raster maps with expert elicitation to generate uncertainty maps for epidemiological studies: Bacterial spot of tomato in Florida as a case study. *Manuscript in preparation*.

PRESENTATIONS

American Society of Tropical Medicine and Hygiene (ASTMH) Annual Meeting Washington, DC | Nov 2026
Mosquito salivary protein antibodies as efficient endpoints for vector control trials: A mechanistic comparison with clinical and entomological outcomes (*accepted oral presentation*).

University of Florida Public Health and Health Professions Research Days (PHHP Days 2026) Gainesville, FL | Apr 2026
A non-homogeneous Poisson process model linking serological biomarkers of Aedes aegypti bite exposure to vector control intervention efficacy (*oral presentation*).

University of Florida Emerging Pathogens Institute Research Days (EPI Days 2026) Gainesville, FL | Apr 2026
Mosquito Bite Exposure Biomarkers as Effective Endpoints for Cluster randomized trials (*poster presentation*).

Summer Institute in Statistics and Modeling in Infectious Diseases (SISMID 2025) Atlanta, GA | Jul 2025
A Unified Computational Framework for Evaluating Vector Control Trial Endpoints: Nonhomogeneous Poisson Modeling and an Interactive R Shiny Implementation (*oral presentation*).

HONORS & AWARDS

- **PhD Travel Award**, American Society of Tropical Medicine and Hygiene (ASTMH), 2026.
- **PhD Student Travel Award**, Summer Institute in Statistics and Modeling in Infectious Diseases (SISMID), 2025.
- **Meritorious Mentor Award**, Mathematical Modeling Competition, China Society for Industrial and Applied Mathematics (CSIAM), 2022.
- **National Scholarship**, Ministry of Education of the People's Republic of China, 2020.
- **Second Prize**, China Undergraduate Mathematical Contest in Modeling (CUMCM), China Society for Industrial and Applied Mathematics (CSIAM), 2015.